

KRISHNA KUMAR

Data Engineer | Python & SQL Specialist | ETL & Cloud Pipelines

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PROFESSIONAL SUMMARY

Data Engineer with 1.5+ years of hands-on production experience designing and automating ETL pipelines processing 5M+ records daily across 4+ enterprise databases (Oracle, Snowflake, SQL Server) and cloud data warehouses (Amazon Redshift) in the banking domain. Proven track record reducing manual effort by up to 90% through Python and PySpark automation, cutting pipeline failures via self-healing architectures, and driving ~35% infrastructure cost savings. Adept at writing complex SQL transformations and deploying production-grade data solutions on AWS—delivering reliable, auditable, and high-performance data infrastructure.

SKILLS

Programming: Python (ETL, automation, data processing), PySpark, SQL (complex queries, stored procedures, window functions)

Cloud & ETL: Databricks, AWS Glue, AWS Lambda, Amazon Redshift, AWS S3, EC2, IAM, Appflow, DBT

Databases: Oracle DB, Snowflake, SQL Server, Amazon Redshift, AWS Athena

Big Data: Apache Spark, Hadoop, Hive, Kafka

DevOps / Monitoring: Docker, CloudWatch, Grafana, Git, GitLab, CI/CD

Web / Other: Django, Streamlit

PROFESSIONAL EXPERIENCE

TCS (Tata Consultancy Services),

Data Engineer 02/2025 – Present | Bengaluru

- Designed and implemented production ETL workflows using Python and SQL, consolidating data from 4+ enterprise sources (SQL Server, Snowflake, Oracle) into Amazon Redshift, processing 5M+ records daily for analytics and reporting.
- Wrote 50+ advanced SQL queries—joins, window functions, CTEs, stored procedures—to transform and validate data at scale, improving query performance by ~30% across enterprise databases.
- Developed 10+ AWS Glue and Lambda functions in Python/PySpark to automate job orchestration and trigger ETL processes, eliminating manual intervention and reducing pipeline runtime by ~40%.
- Built and monitored data synchronization processes, across 20+ pipelines, ensuring 99.5% on-time data delivery; used CloudWatch and Grafana for real-time pipeline health tracking.
- Configured Amazon S3 bucket policies and AWS IAM roles across 15+ buckets to enforce secure data access controls and regulatory compliance.
- Collaborated with analysts, stakeholders, and cloud architects across 3+ cross-functional teams to translate business requirements into scalable, production-grade data solutions.
- Managed incident and change processes—including root cause analysis for data failures—reducing pipeline downtime by ~25% and maintaining runbooks for faster recovery.

Freelance,

Backend Developer / Data Consultant 05/2022 – 12/2024 | Remote

- Built complex data replication and ETL pipelines primarily using Python and AWS Glue/PySpark for 5+ clients in healthcare and hospitality sectors, processing 2M+ records monthly.
- Developed Python automation and monitoring scripts to track resource utilization (disk growth, memory) across 10+ EMR core nodes, reducing capacity-related incidents by ~35%.
- Delivered custom dashboards and applications backed by SQL databases for 8+ clients, providing real-time operational visibility and cutting manual reporting time by ~50%.

PROJECTS

ServiceNow Incident Automation,

SEI / TCS 02/2025 – Present

Tech Stack: Python, SQL (Oracle DB), Streamlit, AWS EC2, ServiceNow API

- Engineered a Python bot integrated with Oracle DB that auto-routes 500+ daily ServiceNow incidents based on real-time shift roster data—reducing manual ticket assignment effort by ~90%.
- Built SQL queries against Oracle DB to fetch live roster and assignment data; deployed a Streamlit monitoring dashboard on AWS EC2 for 24/7 visibility into ticket assignments and availability of 50+ engineers.

SmartETL – Self-Healing & Cost-Intelligent Data Pipeline

05/2026 – Present

Tech Stack: PySpark, AWS EC2/S3, Databricks, Snowflake, Apache Airflow, Claude API, Streamlit

- Architected a self-healing ETL pipeline on AWS EC2 that ingests a 7-table, 58M-row banking dataset from S3, automatically detects 5 failure types (schema drift, null surges, type mismatches), and resolves them via Claude API-generated PySpark fixes in under 30 seconds with zero manual intervention.
- Built a cost intelligence module tracking real-time Spark, S3, and Snowflake spend, using anomaly detection and an LLM-powered optimization engine to surface actionable savings — demonstrating ~35% cost reduction (~\$183/month) in simulated runs.

DataGuard – Automated PII Detection & Compliance Masking Pipeline

11/2025 – 02/2026

Tech Stack: Python, PySpark, AWS S3, Snowflake (Dynamic Data Masking), Apache Airflow, Regex/NER

- Built an automated PII detection & masking pipeline addressing RBI/GDPR data-privacy compliance for banking datasets — using regex + NER models to identify 12+ sensitive data types (PAN, Aadhaar, account numbers) across 50+ columns at 99%+ detection accuracy.
- Integrated Snowflake Dynamic Data Masking with automated audit-trail logging via Airflow, processing 10M+ records and cutting manual compliance review effort by ~80% with zero reported violations.

EDUCATION

LS College, Bihar University — Bachelor of Computer Applications (BCA)

07/2021 – 07/2024 | Bihar

COURSES & TRAINING

- AWS Certified Solutions Architect – Associate (Udemy)
- AWS Certified Data Engineer – Associate (Udemy)